

ARYAN PERSHAD

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Aim: to commercialise stem cell and gene editing technologies to treat CNS ailments (stroke, TBI and epilepsy in particular). I hope to integrate scientific theory and advanced computing with wet lab research to explore and solve longstanding problems in the management of CNS diseases. My history with epilepsy & gut issues lends my work personal relevance.

EDUCATION

AUGUST 2014 TO JUNE 2025: SINGAPORE AMERICAN SCHOOL (SAS), CLASS OF 2025

Cumulative GPA: **4.4 / 4.0**

SAT Score (May 2024): **1550 (Reading and Writing: 780, Math: 770)**

Coursework includes **13 APs**: AP Biology, AP Physics C, AP Calculus BC, AP Seminar, AP Literature, AP Comparative Government and Politics, AP US Government and Politics, AP Macroeconomics, AP Microeconomics, AP English Language, AP U.S. History, AP Physics 1

Selected academic accomplishments (in reverse chronological order):

- Secretary General – SASMUN VII Conference (2024): selected by peers and MUN director to lead the school's annual Model UN conference
- John Locke Institute Public Policy Symposium at Princeton University (Theme of Energy and Environment Policy); selected for program despite being below age cutoff (at 15 years of age); engaged with world leading economists, scientists, and philosophers in discussions on climate policy
- National Merit Semifinalist 2023/2024, **PSAT Score 1500**
- World Scholar's Cup (2023): Singapore Regional Round **#1 individual overall**, Writing; Doha Global Round **#1 team overall**, Writing; qualified for Tournament of Champions at Yale
- John Locke Institute (Oxford & Princeton): **Finalist**, 2022 Global Essay Competition (Junior Category); Received **commendation**
- Model UN (Singapore American School MUN, Autumn 2022): **3x Best Delegate Award** recipient
- Regional **finalist** (team): 2021 National History Day competition (project: Cuban Missile Crisis and importance of communication)
- MATHCOUNTS 2020 (7th grade): Member of school's travelling team (**top 10 in school**); individually qualified for regional final in Indonesia

MAY 2020 TO MAY 2024: CRIMSON GLOBAL ACADEMY (CGA), IGCSE AND ADVANCED COURSEWORK

While enrolled as a full-time eighth-grade student at SAS, prepared for and completed 7 IGCSE exams; results:

Physics: **High A-star** (9 on 1 to 9 scale) Math: **High A-star** (9) Economics: **High A-star** (9) Business: **High A-star** (9)
Chemistry: **A-star** (8) Biology: **A-star** (8) History: **A-star** (8)

N.B.: CGA faculty recommends that students already enrolled in full-time schools take no more than 1 CGA course

Beginning in September 2021, began two customised, multi-year courses: (1) **mathematics for machine learning and quantum computation** (topics include linear algebra, analytic geometry, vector calculus and statistics) and (2) **A-Level physics**

February 2022: Published neural network simulation: <https://ajd.itch.io/learning-2-learn>

JUNE 2019: HARVARD UNIVERSITY THREE-PART ONLINE COURSE IN NEUROSCIENCE (EDX); GRADE: PASS

Completed "Fundamentals of Neuroscience" parts 1, 2 & 3 during the summer after 6th grade (age 11; scores: **82%, 91% & 89%**)

AUGUST 2012 TO JUNE 2014, HONG KONG INTERNATIONAL SCHOOL (HKIS)

SEPTEMBER 2009 TO JUNE 2012, FOREST BLUFF MONTESSORI (LAKE BLUFF, ILLINOIS)

EXPERIENCE

JULY - AUGUST 2025: E-THERAPEUTICS

Returned to e-therapeutics (now Tangram Therapeutics) for a longer, 7-week stint as the primary prompt engineer on the AI team. Developed LLibra-Prompt, an evolutionary LLM-as-a-judge prompt engineering algorithm from idea to product stage available company-wide.

JUNE – AUGUST 2024: JOM MAGAZINE

Collaborating with the media team at Jom Magazine (<https://www.jom.media/>) to publish an article detailing personal experience with epilepsy and lessons for families with epileptic children (To be published August 9th 2024)

JULY 2024: E-THERAPEUTICS

Returning to company headquarters for three weeks to build upon experience from the summer of 2023 as detailed below

First ever two-time intern at the company

JUNE 2024: HARVARD UNIVERSITY, HARVARD STEM CELL INSTITUTE (HSCI)

Returned for a three-week internship in the lab of Dr. Jeffrey D. Macklis (<https://hsci.harvard.edu/people/jeffrey-d-macklis-md>)

First ever two-time intern at the Macklis Lab, understanding is that the same is true for HSCI

Applied skills learned in 2022 to conduct immunohistochemistry experiments, including tissue staining, epifluorescence microscopy and imaging; applied experience from e-therapeutics to analyse data as part of a bioinformatics project run by Dr. John Froberg and Harvard graduate student Dustin Tillman

JULY 2023: E-THERAPEUTICS

Three-week internship in the company's London headquarters (invited to return in 2024 by Ali Mortazavi, company CEO)

Youngest ever intern at the company (age: 15)

Worked on Large Language Models in collaboration with team software engineers, developed a proteomics machine learning algorithm to predict protein structure based on amino acid sequence as part of the Human Proteome Project

NOVEMBER 2022 – PRESENT: MEDANTA – THE MEDICITY (GURUGRAM, INDIA)

Consulted with Dr. Atma Ram Bansal (Associate Director, Epilepsy Program) to design outreach programs for pediatric epilepsy patients in local community; focus: increasing awareness of treatment options & post-seizure lifestyle management; implemented these programs on a 1 on 1 basis

JUNE – JULY 2022: HARVARD UNIVERSITY, HARVARD STEM CELL INSTITUTE

Three-week internship in the lab of Dr. Jeffrey D. Macklis (<https://hsci.harvard.edu/people/jeffrey-d-macklis-md>)

Youngest ever intern at HSCI (age: 14), invited back to return before completing high school

Shadowed scientists and lab technicians; participated in weekly lab meetings; learned PCR genotyping, tissue prep & sectioning, immunolabeling, epifluorescence & confocal microscopy, electroporation

JULY 2021 – PRESENT: CAD TEAM, OUTREACH LEAD, FIRST ROBOTICS COMPETITION (FRC); HEAD MENTOR, MIDDLE SCHOOL ROBOTICS (VEXIQ)

Specialty: CAD (computer-aided design); selected by 12th grade captains to join the most competitive robotics team at SAS, served as Outreach lead for one year; SAS FRC team had **best performance to date** in 2024 @ Sydney; mentors previously participated in middle school robotics program

JUNE 2019 – JUNE 2021: TEAM LEADER, VEX IQ ROBOTICS TEAM

Specialty: CAD (computer-aided design); also deeply involved in robot building and training for autonomous runs

Won Design Award at World Championship 2021 (second-highest award at the Guinness World Record-certified largest online robot championship in the world; 12,693 participants from >30 countries; first such award for SAS)

INTERESTS

Entrepreneurship; Robotics (competing/mentoring); writing; composing music & poetry; grand strategy games; investing; baseball; Model United Nations